

# SQL Metadata Guide

This document provides business metadata used by the ESG Natural Language to SQL Generator. It describes analytical entities, dimensions, metrics, aggregation rules, business synonyms, and SQL generation guidance. It is intended to be indexed in Chroma and retrieved during SQL generation.

## Primary Dataset

All analytical queries target the table `esg_data`. Only read-only `SELECT` statements are supported.

## Business Entities

Facility, Country, City, Industry and Fiscal Year are business entities used for grouping, filtering and drill-down.

## Dimensions

Facility\_Name, Country, City, Industry and Fiscal\_Year are dimensions. Use dimensions in `GROUP BY` clauses. Multiple dimensions may be combined when explicitly requested.

## Metrics

Scope1\_Emissions, Scope2\_Emissions, Scope3\_Emissions, Energy\_Consumption\_MWh, Renewable\_Energy\_Percentage, Water-Withdrawal\_ML, Water\_Recycled\_Percentage, Waste\_Generated\_Tons, Waste\_Recycled\_Percentage, Employee\_Count, Female\_Workforce\_Percentage, Training\_Hours and Compliance\_Violations are analytical metrics.

## Aggregation Rules

`SUM` → Emissions, Energy, Water Withdrawal, Waste Generated, Employee Count, Compliance Violations.

`AVG` → Renewable Energy %, Water Recycled %, Waste Recycled %, Female Workforce %, Training Hours.

`COUNT` → Number of Facilities or Records.

## Sorting Rules

Highest / Top → `ORDER BY` metric `DESC`.

Lowest / Bottom → `ORDER BY` metric `ASC`.

Rankings should always include `LIMIT` when the user specifies Top N or Bottom N.

## Trend Analysis

Trend questions should group by `Fiscal_Year` and sort `Fiscal_Year` ascending.

## Comparison Analysis

Comparison questions should `GROUP BY` the requested business dimension such as Country, Industry or Facility.

## Drill-down Hierarchy

Global → Country → City → Industry → Facility.  
Preserve existing filters unless the user explicitly replaces them.

## Business Synonyms

Carbon Footprint → Scope1 + Scope2 + Scope3 Emissions.  
Carbon Emissions → Scope Emissions.  
Green Energy → Renewable\_Energy\_Percentage.  
Women Representation / Gender Diversity → Female\_Workforce\_Percentage.  
Headcount / Workforce → Employee\_Count.  
Plant / Site → Facility\_Name.

## Filtering Guidance

Country filters apply to Country.  
Industry filters apply to Industry.  
Fiscal year references apply to Fiscal\_Year.  
Facility names should filter Facility\_Name.

## Visualization Mapping

Trend → Line Chart.  
Comparison → Bar Chart.  
Ranking → Horizontal Bar Chart.  
Distribution / Share → Pie Chart.  
Single KPI → Metric Card.  
Correlation → Scatter Plot.

## Unsupported Operations

Never generate INSERT, UPDATE, DELETE, DROP, ALTER, CREATE, MERGE, TRUNCATE, PRAGMA, ATTACH or DETACH. Only SELECT statements are permitted.

## Executive Reporting Guidance

Executive summaries should explain business findings, highlight significant comparisons, identify highest and lowest values where meaningful, discuss observable trends and avoid SQL terminology.